

Design of an Exercise Game for Older Adults: Promoting Cognitive Health through Combined Aerobic Exercise and Foreign Language Learning

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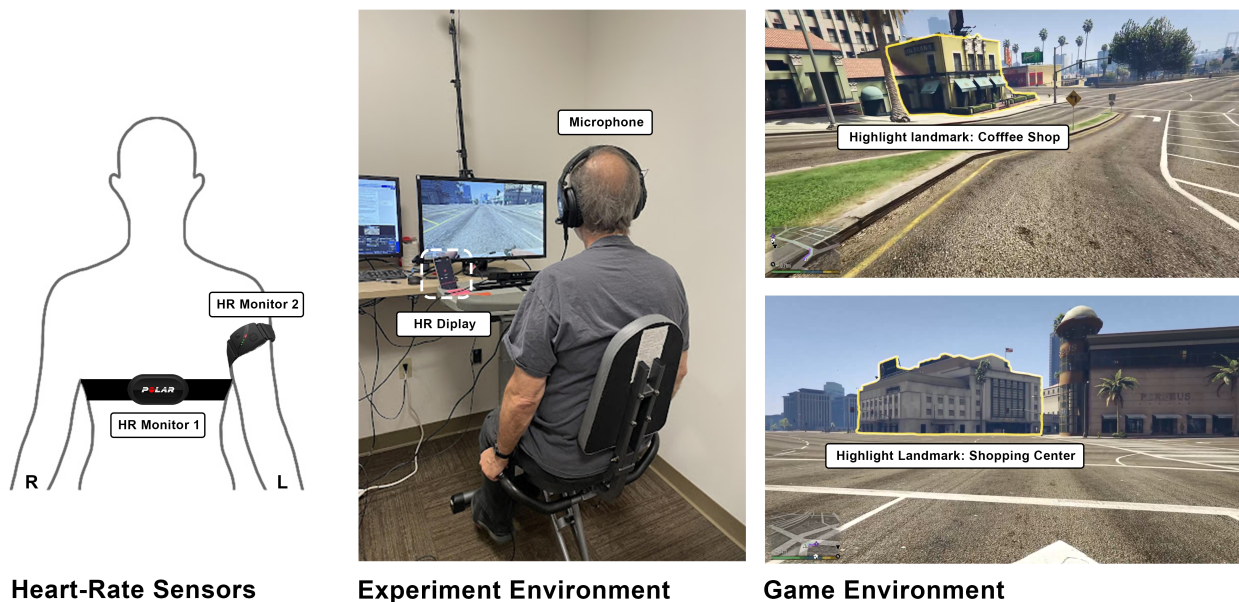


Figure 1: A senior participant in the combined aerobic exercise and language learning group, 2023

ABSTRACT

Aging is a primary contributor to cognitive decline. Existing research demonstrates that Aerobic Exercise (AE) induces significant

functional neuroplasticity changes, enhancing cognitive function. Foreign language learning (FLL) also positively impacts cognitive health and may offer resistance to age-related cognitive decline. Addressing the gap in studying the combined effects of AE and FLL on cognitive benefits, we designed an intervention using a screen-based virtual biking tour to enhance cognitive health in older adults. Our project investigates whether the simultaneous integration of AE and FLL yields superior cognitive and language learning outcomes compared to FLL learning alone. In a preliminary analysis with a small sample, we observed significantly improved Spanish language learning outcomes in both combined physical

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and language learning groups. The significant cognitive function improvements are also observed in the FLL group after a short-term language learning.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Cognitive Health, Exercise Game, Learning, Older Adults

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1 INTRODUCTION

Maintaining one's cognitive health span is an essential component of healthy aging. Several lifestyle interventions have demonstrated efficacy in improving functional cognitive performance in older adults. These include aerobic exercise (AE), and cognitive training (CT), such as foreign language learning (FLL). Extensive research has consistently shown that AE leads to significant structural and functional neuroplasticity resulting in enhanced cognitive functions [6, 9, 11] and an improved sense of overall well-being [3]. The findings from the effects of CT programs, on the other hand, show mixed results. Cognitive training programs, while effective at improving the specific skills targeted in training, often do not result in meaningful transfer to untrained tasks [20, 21]. One potential reason for this lack of transfer may be the relative specificity of most types of CT. FLL is a task that demands a broad array of skills, including attention [30] and working memory [33]. Neuroimaging research has connected improvements in brain structure in young adults and functional brain changes in older adults to learn a foreign language [4, 17]. Furthermore, studies have suggested that long-term FLL may provide resistance to some aspects of cognitive aging [1, 2, 8].

It is worth noting that technology is commonly employed to facilitate cognitive training (CT). Existing computer-based CT programs have demonstrated their effectiveness in enhancing reasoning skills [7], short-term memory [18], working memory [13, 23, 28], and processing speed [31] in older individuals. However, the research on the use of technology in FLL by healthy elderly individuals is limited. As of 2020, only two studies had examined technology-based language learning in older adults [32, 35]: Both programs improved cognitive functions in older adults, and the benefits were sustained for up to three months after the programs [35].

Several investigations have explored the combination of AE and FLL in young adults, demonstrating that moderate exercise facilitates learning [22]. Another study examined the temporal aspects of exercise and its influence on augmenting long-term memory [10]. Given the limited amount of research focusing on older adults, we present an accessible and novel approach that uses exercise to

enhance foreign language acquisition in this population and potentially promote long-term cognitive benefits. Many previous studies within the human-computer interaction field have examined the preferences and challenges faced by older adults during exercise, travel, and studying foreign languages. Researchers have recommended designing game-related programs for older adults and individuals requiring additional support by simplifying game objectives, providing interactive systems [5] and timely feedback [14] to support learning, and incorporating appropriate visual complexity [29]. It is worth mentioning that we adjusted the program and the learning content based on our participants' preferences and reports of their user experience [24]. In our design, we have a) reduced the visual complexity for older individuals by adopting a purely auditory learning method and b) provided timely feedback to older adults while they were practicing.

To explore the feasibility and effectiveness of a newly developed training paradigm, we created an accessible and immersive stationary virtual bike tour in Castilian Spanish. The tour is divided into episodes utilizing the virtual environment and incorporates exercise game elements to promote participants' enthusiasm. Our approach provides immersive virtual biking visuals and auditory information while monitoring participants' heart rates to ensure they are engaged in moderate exercise. The tour scripts include Spanish phrases spoken by a "biking buddy" who prompts participants to repeat after him. Our learning methodology is rooted in the understanding that older learners can absorb information about a foreign language even without direct teaching, and that they are capable of extracting linguistic "rules" implicitly, via pattern recognition.

This learning environment is designed to mimic many features of how children acquire their first language in that it contains no written text, relies entirely on auditory processing, and conveys information only by the learner hearing speakers using the same structures repeatedly. Importantly, this method has been demonstrated to be a feasible way for adults to extract and acquire syntactical patterns from artificial language [15, 25].

We believe that designing the computer-based, auditory language learning paradigm will provide us with a tool to investigate several areas of interest. First, it will allow us to bridge the gap in understanding auditory-based FLL in older adults. Second, we can explore the effects of exercise on auditory-based FLL. Lastly, we can examine the impacts of combining AE and FLL strategies on improving cognitive function in older adults. While we may not be able to address all these questions simultaneously, the primary goal of our current pilot study is to assess the feasibility of this combined paradigm in foreign language learning outcomes and to observe whether short-term FLL learning has any effects on cognitive function performance in older adults.

We tested language learning efficacy after four 30-minute sessions on 12 older adults who had minimal exposure to Spanish (6 with exercise, 6 without) recruited from the greater Boston community. After a preliminary analysis with a small sample, we found that Spanish learning outcomes were significantly improved in both combined physical and language learning groups and single language learning groups after four sessions of training. We also observed that attention and processing speed improved in participants after short-term foreign language learning, especially in the

Language learning-only group (FLL group). However, only the exercise and language learning group (AE + FLL) showed a trend of increased working memory performance. This preliminary result suggests a promising approach to implementing auditory-based Spanish Learning for foreign language study and cognitive training and creating a new structure for future exercise game development that aims to promote the health of older adults comprehensively.

2 VIRTUAL ENVIRONMENT DESIGN

2.1 System Overview

The virtual system was developed by modifying the open-world game Grand Theft Auto 5 (GTA 5) using the C# programming language. We selected GTA 5 due to its extensive and detailed geographical content, offering high-quality representation of landmarks. Third-party mods, including gta5-real-mod, were incorporated to create a realistic 2D virtual city bike tour for participants. The narrative script and Spanish audio were generated using the Voicemaker website. We recorded footage from GTA 5 and then edited this in Adobe Premiere to incorporate AI-generated audio and visual cues of landmark highlights.

2.2 Spanish Learning in Game

Targeted Spanish knowledge was carefully selected for learners with minimal exposure to Spanish and categorized into the recognition of nouns and daily phrases, as well as several syntactical patterns in Spanish that differ from English: gender agreement with article, reversal of adjectival agreement with noun-gender, use of prepositions, and some basic contrasts involving pronouns/conjugations. During the game, participants' exposure to Spanish was entirely auditory-based with no visual text display. The structure of the Spanish was highly repetitive and used contrasting pairs to help learners identify patterns.

Each phrase was presented to participants in brief chunks of a few words at a time, which they were encouraged to repeat immediately, and they would then hear the full sentence again. After they had heard each target phrase with a particular noun or noun phrase (e.g., "la iglesia") twice, they would be prompted with a one-question comprehension quiz to receive some feedback on the patterns they were extracting, see Figure 2.

2.3 Spanish Learning Opportunities

The experiment consists of four 30-minute cumulative virtual bike routes. Three non-playing characters (NPCs) accompany the participant on the tour, see Figure 2.

- Joshua (Narrator): A male native English speaker who acts as the tour guide in the game. He introduces landmarks to participants and gives detailed descriptions of landmarks in English, but with no direct correspondence to the Spanish, except in the name of the landmark.
- Lorenzo ("Bike Buddy"): A male native Castilian Spanish speaker who acts as the local Spanish tour guide as well as a Spanish instructor in the game. He speaks Spanish phrases to participants according to which landmarks are coming up and directs participants to repeat after him to learn.

- Ashley (Companion): A female native English speaker who acts as the travel companion who is trying to learn Spanish in the game. She frequently requests that Lorenzo repeat Spanish sentences for participants. She also helps participants to engage actively in the learning process by re-stating parts of Lorenzo's phrases to "make sure she is understanding." These statements provide the context of the in-game mini-quiz for participants, and they involve examples of English-accented Spanish.

In our approach, participants extract the meaning of the Spanish phrases by listening to cues from the English narration, observing the environment, such as highlighting landmarks, and noting what parts of the Spanish phrases are consistently repeated and which are not, relative to events on the tour. Each 30-minute virtual bike route includes 10 landmarks, such as churches, banks, or golf courses. At each landmark, a highlight appears to guide participants' attention to that landmark. The goal is to help learners recognize patterns in Spanish vocabulary as they listen and engage with the virtual tour.

Characteristics	AE + FLL (N=6)	FLL (N=6)	p-value
Age	M=72.33 SD=2.8	M=76.23, SD=5.04	0.129
MoCA	M=26.66 SD=1.96	M=26.66 SD=1.86	1.000
KBIT	M=113.83 SD=12.73	M=118 SD=10.65	0.553
Pre-Spanish	M=0.42 SD=0.22	M=0.36 SD=0.16	0.711
Gender	Female=1 Male=5	Female=2 Male=4	N/A

Table 1: The variation of Age, MoCA, KBIT, and Existing Spanish Knowledge and Gender between AE + FLL and FLL groups.

3 METHOD

3.1 Participant Recruitment

We recruited participants through posters in the area around the University, senior centers, and bulletin boards in the local community, and also through email and phone calls. Potential participants were evaluated for eligibility through phone screening or online screening survey questionnaires. All participants were generally healthy, without significant physical impairments, visual, or hearing acuity issues. Those with a history of neurological diseases (Alzheimer's, Parkinson's, epilepsy, chronic migraines, multiple sclerosis, stroke) or heart-related conditions (unstable angina, heart attack, heart failure, arrhythmia, valve diseases) were excluded. People with high blood pressure with medication were included. All participants had minimal Spanish exposure, with no formal Spanish learning experience. Participants who regularly had more than 60 minutes of moderate exercise, such as cycling and jogging, per week in the past 6 months were excluded. We used standard cut-off scores for older adults on the Montreal Cognitive Assessment (MoCA ≥ 24) and Kaufman Brief Intelligence Test 2nd Edition (KBIT-2 ≥ 85) to ensure that participants were qualified for the integral cognitive and intellectual demands of our experiments.

Participants were randomly assigned to the biking language learning group (AE + FLL Group, N=6) or language learning only group (FLL Group, N=6) for a total of twelve participants (3 female, 9

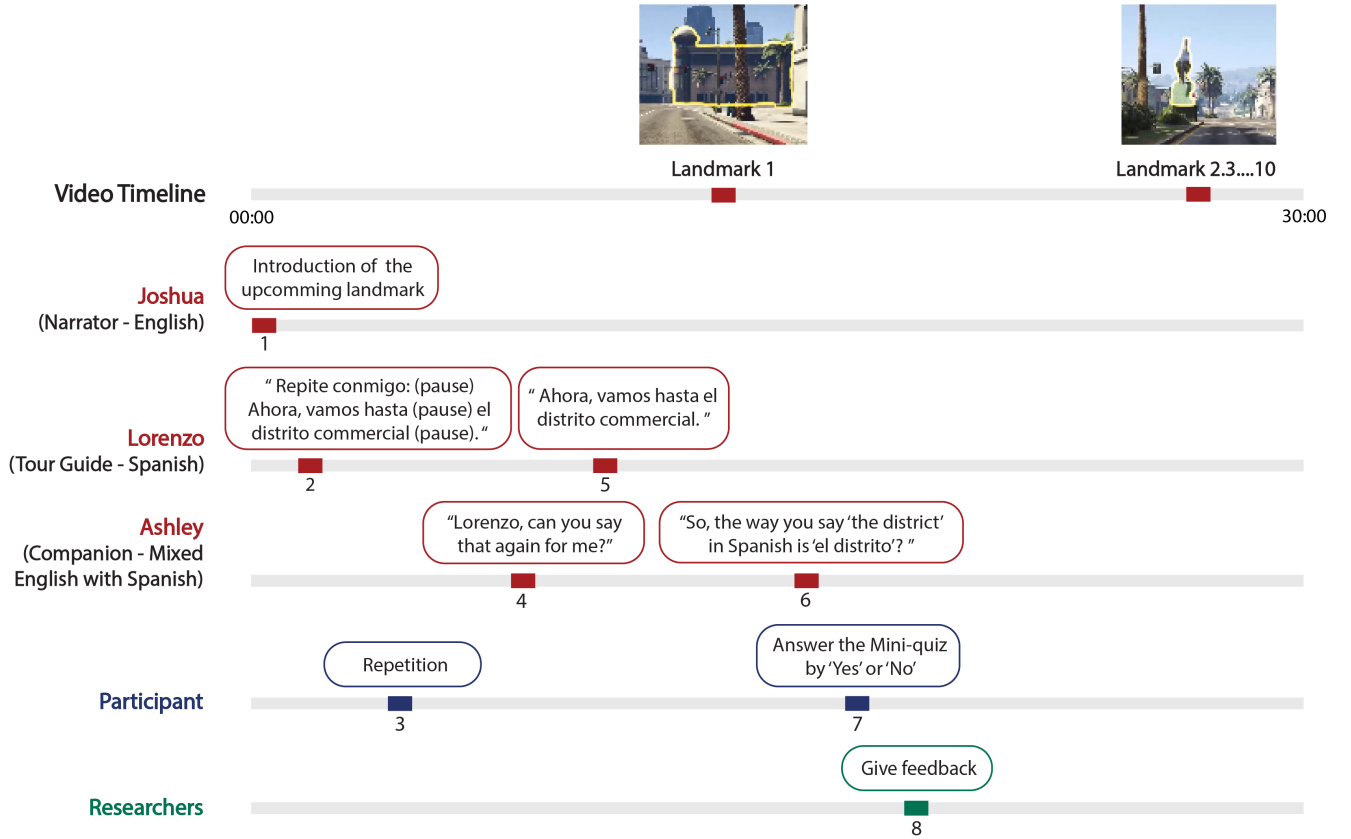


Figure 2: The Spanish learning process in the game involves interactions among game characters (NPCs) providing language learning cues, participants, and researchers offering feedback. The process follows the steps marked with numbers from one to eight.

male) aged 65 to 85 ($M=74.28$, $SD=4.39$). We observed no significant differences between age, global cognitive functions (MoCA), IQ (KBIT), and existing Spanish knowledge between groups at the start of the study (Table 1).

3.2 Experiment Environment Set-up

All experimental sessions for each participant were conducted in the same laboratory setting. The setup included a stationary bike, a LG mobile phone, two computer monitors, a Polar H10 chest HR monitor, a Rhythm 24 Scosche armband HR monitor, and an EyeTech VT Mini3 eye-tracking device (no eye movement data is presented in this manuscript). Lab Streaming Layer (LSL) was used to synchronize Polar H10 raw ECG data with EyeTech VT Mini3 eye-tracking data for future analyses. The Elite HRV application was used on the LG phone to display the heart rate data for participants from the Scosche Rhythm 24 Armband. Elite HRV was operated on a LG phone and was placed within the participants' line of sight for them to monitor their own heart rate in real-time. OBS studio and PsychoPy were used to record original experimental video and audio as an archive for researchers to analyze player engagement and learning processes.

3.3 Procedures

The complete experiment involves six visits over approximately eighteen days. During the initial 90-120 minute visit, eligible participants had their blood pressure measured, received a detailed explanation of the experimental procedures, and signed consent forms. A demographic survey was conducted to collect information on age, gender, education, marital status, and income. Participants also took a pre-test of their Spanish knowledge.

After the first visit, participants were randomly assigned to either the AE + FLL Group (biking and language learning) or the FLL Group (language learning only). For the 2nd to 5th visits, participants engaged in auditory-based Spanish learning, with or without stationary biking. Researchers provided heart rate monitors, adjusted bike seats, and calculated the target heart rate range based on age (64% to 76% of $(220 - \text{age})$). In the AE + FLL group, participants pedaled while maintaining their heart rate within the target range for 30 minutes. A researcher supervised each session, handling equipment, monitoring for emergencies, and providing feedback during mini-quizzes. Each of the 2nd to 5th visits lasted 60-90 minutes. After the 5th visit, participants were invited back to the lab

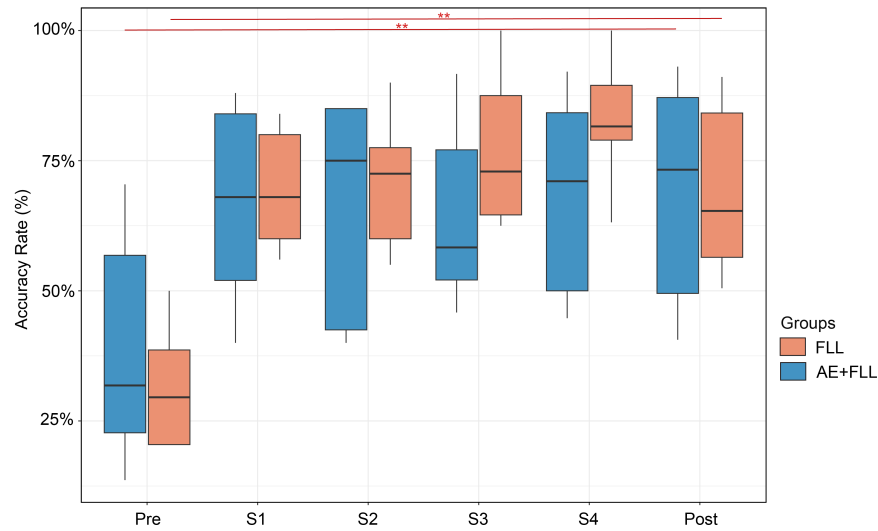


Figure 3: Participants' Spanish learning performance in pre-training, after training sessions (S1-S4), and post-training. Both pre- and post include questions on nouns, noun gender, adjective order, prepositions, and pronouns. S1 focuses on the teaching of nouns and noun gender-article gender agreement; S2 focuses on nouns, adjective order, and adjectival agreement with nouns based on gender; S3 focuses on nouns and prepositions; S4 focuses on nouns, pronouns, and verbs. Sessions are cumulative.

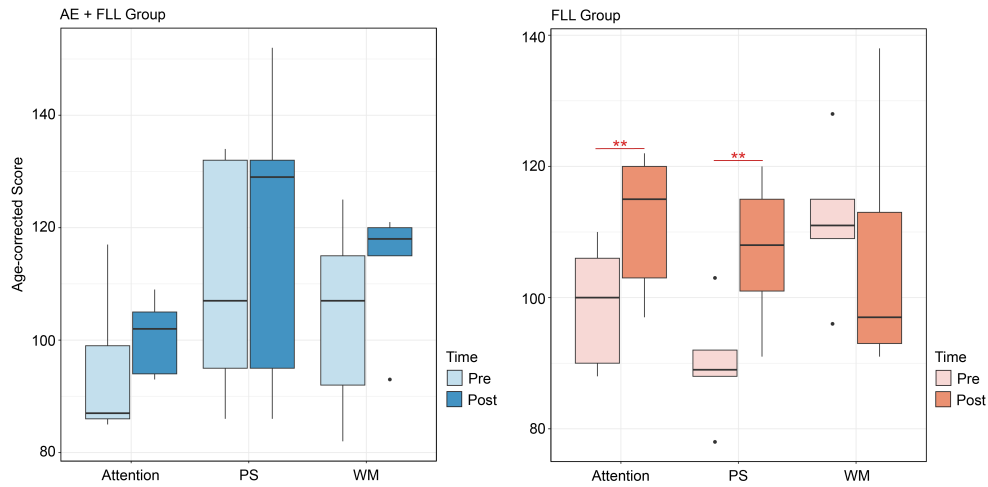


Figure 4: Pre- versus post-test cognitive scores for participants in the AE + FLL group and LL group. The figure on the right shows the same comparison for participants in the FLL group. 'PS' stands for 'Processing Speed'; 'WM' stands for 'Working Memory'.

for a 6th visit to complete another round of cognitive tests and the Spanish knowledge post-test.

3.4 Data Collection

Participants' accuracy rates on the Spanish pre- and post-tests were collected. The tests include a variety of vocabulary terms and probed each syntactical function that would be the learning objective of a lesson during the experiment. The pre- and post-tests follow a similar format, with adjusted question prompts to

minimize exposure bias. Both tests assess the same language knowledge, covering questions on nouns, noun genders, adjective order, prepositions, and pronouns. The accuracy rate from the pre-test is used as a baseline to compare with their post-test accuracy rate. The post-test directly measures the language (both vocabulary targets and syntactical rules) to which the participants were exposed. Participants' demographic information, such as age, gender, marriage status, education level, and income level were collected. We

also collected participants' language history through The Language History Questionnaire (LHQ).

A series of tests measuring cognitive capacities were also administered including the Montreal Cognitive Assessment (MoCA) and Kaufman Brief Intelligence Test Second Edition (KBIT-2), and cognitive battery of tests from the NIH toolbox [12] including the Flanker Inhibitory Control and Attention Test (Attention), List Sorting Working Memory Test (WM), Pattern Comparison Processing Speed Test (PS). The cognitive battery scores are all age-corrected and based on norm-referenced, standardized tests used frequently in research and clinical practice. The pre-training cognitive function tests (Attention, WM, and PS) were used as a baseline to compare the post-training cognitive performance to assess potential improvement.

4 RESULTS

4.1 Language Learning Performance and Retention

A paired sample t-test was used to assess increases between participants' pre- and post-test accuracy for both groups. We found that Spanish learning outcomes were significantly improved in both the AE + FLL group ($t = -6.483$, $df = 5$, $p = 0.001^{**}$) and the FLL group ($t = -5.876$, $df = 5$, $p = 0.002^{**}$) after four sessions of training, see Figure 3.

We examined potential differences in participants' pre-test and post-test rates as a function of group using independent samples t-test. We did not find significant differences between groups' pre-test accuracy rates ($p = 0.712$), indicating a similar baseline level for each group. We also found no significant differences between the two groups' post-test performance ($p = 0.877$).

4.2 Cognitive Task Performance

A paired t-test was used to conduct a within-group analysis in comparing the pre- and post-test performances in three NIH Toolbox tasks: Attention, Processing Speed (PS), and Working Memory (WM) tests. Both groups show the trend of increased cognitive performance after four sessions but only the FLL group shows the statistical differences in the Attention test ($t = -5.6429$, $df = 5$, $p\text{-value} = 0.002^{**}$) and PS test ($t = -5.6352$, $df = 5$, $p\text{-value} = 0.002^{**}$), see Figure 4.

5 DISCUSSION

We described the design and preliminary data for a novel paradigm combining exercise and an audio-based language learning program. Differing from traditional text-based language learning, we employed a "Listen and Repeat" approach with audio-based learning inspired by the way children acquire their first language. Our preliminary results show learning improvements regardless of whether participants' language learning was paired with exercise. This result suggests that the combination of aerobic physical exercise and auditory-based foreign language learning may be a viable offering for older adults seeking to improve their physical and cognitive fitness.

There has not been extensive study on learning during exercise; however, some studies suggest that the combination can enhance learning outcomes [27]. Given the potential for cognitive overload

of the particular combination of learning and exercise we used, we are heartened to see statistically equivalent learning across both groups. Other results, albeit from studies that use more standard cognitive training paradigms [16, 19, 26, 34], suggest that over longer periods of training the neuroplasticity boost wrought by aerobic physical activity will promote better learning.

Our current sample size is too small to draw any firm conclusions. However, we did observe some trends in the data that seem reasonable at face value. Consistent with previous research on exercise adding benefits to executive functions, we observed an increase in performance in the AE + FLL group on the Flanker and List Sort Working Memory tasks [23, 28]. Both groups showed increased attention scores after training. However, only the AE + FLL group showed an improvement post-training in our working memory task. So far, only the FLL group has shown significant improvements in attention and processing speed.

If the findings, such as the enhanced FLL learning outcomes and cognitive functions, remain consistent in a larger-scale study, the inclusion of aerobic exercise alongside a cognitive training program could potentially offer older adults additional advantages. These benefits might not only pertain to their anticipated physical health benefits but could also extend to increased engagement and motivation.

6 CONCLUSION AND FUTURE

We created an accessible and immersive stationary virtual bike tour in Castilian Spanish to explore the feasibility and effectiveness of a newly developed training paradigm that combines exercise and language learning for older adults. Our preliminary results suggest that a program like this can improve older adults' Spanish learning performance. Additionally, language learning itself, with or without exercise, improves cognitive function. However, only the group that includes exercise shows benefits in working memory. Given the constraints of our limited sample size, we will continue to collect additional data to make a conclusive determination regarding the efficacy of the combined AE + FLL condition compared to FLL alone. Future studies will include additional measurements, such as fNIRs, to assess participants' brain activity or changes in brain function within each group during the learning process. This expanded approach will provide a deeper insight into the underlying mechanisms responsible for the observed learning outcomes and cognitive improvements. Without counting the additional benefits from physical activity, a couple of participants in a combined group verbally appreciated the fun they got from virtual biking and language learning. User experience research will be necessary to study the motivation and engagement from combined conditions compared to solo learning. This combined strategy may provide a potential direction for engaging older adults in motor-cognitive activities.

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